

DUET: Endogenous Failure Detection and Recovery via Dual-Action Manifold Alignment

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Abstract—Vision-language-action (VLA) policies map visual observations and language instructions to robot actions. With limited robot post-training data, they can drift from successful behavior during closed-loop execution. We study recoverable execution deviations, where retrying the demonstrated behavior remains feasible, rather than severe failures that require task-level replanning or a new manipulation strategy. Existing solutions often add external failure detectors and collect large amounts of human takeover data, making the resulting systems costly and loosely coupled. We introduce DUET (Dual-manifold Unified Execution and Transfer), a dual-action-head module that predicts both relative actions for smooth nominal control and absolute waypoints for trajectory re-entry. The endogenous disagreement between these action spaces detects when execution has become inconsistent with the demonstrated behavior, while the absolute waypoint-induced correction initiates a retry. Asymmetric training further makes the relative branch sensitive to deviations and the absolute branch robust to visual disturbances. We evaluate DUET in LIBERO simulation and real-world manipulation under recoverable visual, viewpoint, scene, and action disturbances. DUET preserves nominal success (TODO% vs. TODO% for the relative-only baseline) while achieving TODO% retry success, outperforming the strongest baseline by TODO percentage points.

I. INTRODUCTION

Vision-language-action (VLA) policies adapt pretrained vision-language models to map visual observations and language instructions directly to robot actions [1], [2], [3]. They are attractive because one policy can connect semantic instructions to continuous manipulation. Yet a policy trained with a small amount of robot data can leave the support of successful behavior during execution. We refer to this condition as an execution failure: the observation, robot-object configuration, or induced trajectory becomes out of distribution (OOD) with respect to the successful demonstrations. The cause may be an off-center grasp, a reachable object displacement, or a partial object slip, but it may also be external, such as an unexpected occlusion, viewpoint change, or displacement of the object or robot arm. In these cases, the policy may continue issuing locally plausible commands without recognizing that the current rollout should be retried. Detecting such deviations and returning to a demonstrated execution path can improve robustness without requiring a new task planner.

Mainstream approaches commonly introduce an external monitor, critic, verifier, or predictive model for failure detection [4]. When recovery is also required, they typically collect large amounts of human takeover data in which an

operator intervenes and demonstrates a corrective action [5], [6], [7]. This creates substantial data and deployment cost, especially when the disturbances span visual observations, object configurations, and robot poses. More importantly, the detector only indicates that execution has gone wrong, while the recovery controller is trained as a separate capability; their decisions need not be consistent. We ask whether the policy can use its own action predictions to recognize a recoverable OOD deviation and retry the learned behavior without additional failure labels or takeover demonstrations.

DUET is motivated by a simple asymmetry that we observed in preliminary experiments. Relative-action policies produce smooth and stable motion around the demonstrated trajectory, but once a grasp or pose deviates, a local command provides no explicit target for returning to the previous behavior. Absolute-action policies are less suited to fine local control, yet their predicted waypoints can point back toward a reachable portion of the demonstrated trajectory and make a retry possible. DUET therefore learns the same successful behavior in two action spaces: a relative head for nominal execution and an absolute head for trajectory re-entry. We use the disagreement between their predictions as an endogenous signal of OOD execution. The disagreement is expected to remain small when both heads recognize the same in-distribution behavior and to increase when the current rollout is no longer compatible with that behavior.

At inference time, DUET converts the absolute waypoint into a correction from the live pose and compares it with the relative action. A soft gate keeps relative control in the nominal regime and shifts toward the waypoint-induced correction when the disagreement indicates a recoverable deviation, allowing the robot to re-approach the demonstrated behavior and retry. The gate returns to relative control after the two predictions realign. This is a failure-aware retry mechanism, not a general recovery planner: DUET does not claim to solve irreversible failures, unreachable dropped objects, hardware faults, or cases that require discovering a new task strategy.

Our contributions are threefold:

- We formulate failure-aware retry as detecting recoverable OOD execution deviations and re-entering the support of successful behavior, separating this problem from severe failure recovery and task-level replanning.
- We propose DUET, a dual-action-head VLA that uses endogenous disagreement between relative and absolute action spaces to detect deviations and to activate an absolute waypoint-based retry while preserving smooth nominal control.

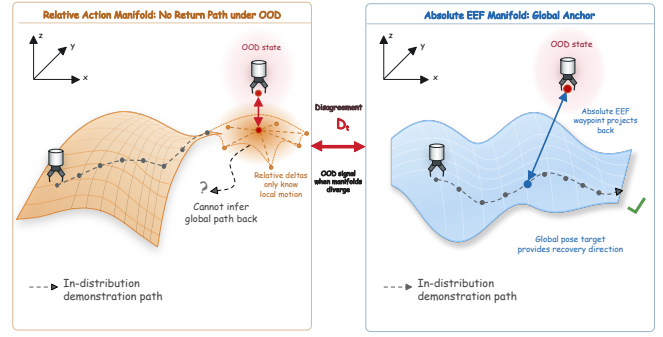
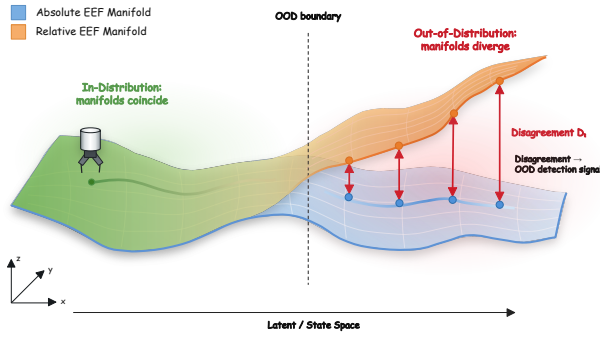


Fig. 1. Failure-aware retry with dual action spaces. Relative and absolute predictions agree during nominal execution but diverge when a recoverable deviation drives the rollout OOD, exposing an endogenous disagreement score D_t . Relative actions provide precise local motion, whereas absolute waypoints bring the robot back toward a demonstrated execution path for retry.

- We evaluate DUET in simulation and on a real robot across diverse visual, viewpoint, scene, and action disturbances, measuring both nominal robustness and retry success without additional human takeover data.

II. RELATED WORK

A. Failure Detection for Robot Policies

Runtime failure detection estimates whether a policy remains within the support of successful behavior. Diff-DAGger uses diffusion uncertainty to request expert intervention [8], while Sentinel monitors action inconsistency and task progress [9]. FAIL-Detect formulates failure detection as sequential OOD testing using only successful executions [4]; FIPER similarly combines representation novelty with action uncertainty without failure data [10].

VLA-specific methods use internal features or external foundation models. SAFE learns a detector from successful and failed rollouts [11], whereas AHA trains a VLM to detect and diagnose failures [12]. Such methods require additional supervision or a separate detector whose output is not a recovery action. DUET instead detects failures through disagreement between two action spaces learned within the policy, using OOD manifold inconsistency as both an alarm and a recovery signal.

B. Recovery from Execution Failures

Recovery methods commonly expand the training distribution with expert corrections. DAGger queries experts on policy-induced states [5]; intervention-based methods collect human takeovers [13], [6]; and IntervGen transforms limited interventions into corrective trajectories [7]. These approaches reduce compounding errors but still require additional recovery actions.

Foundation-model systems instead use semantic supervisors. RACER learns from language-annotated recovery trajectories [14], while AHA and VLM controllers diagnose failures and guide downstream correction [12], [15]. Their detection and recovery components remain separately trained or prompted. DUET couples both capabilities without extra recovery data: absolute states from successful demonstrations provide recovery anchors, while cross-space disagreement determines when and how strongly to move toward them.

III. METHOD

A. Problem Formulation

At time t , the robot observes multi-view images o_t , a language instruction l , and its current proprioceptive state s_t . DUET is trained only on successful demonstrations and predicts the demonstrated behavior in two action representations over a horizon K :

$$\hat{\Delta}A_t = f_{\text{rel}}(o_t, l, s_t) \in \mathbb{R}^{K \times 7}, \quad (1)$$

$$\hat{W}_t = f_{\text{abs}}(o_t, l) \in \mathbb{R}^{K \times 7}. \quad (2)$$

Here, ΔA_t contains relative operational-space-control commands and W_t contains future absolute end-effector waypoints $[x, y, z, r_x, r_y, r_z, g]$. We decompose each command as $a_t = [a_{p,t}, a_{g,t}]$, where $a_{p,t} \in \mathbb{R}^6$ controls end-effector pose and $a_{g,t} \in \mathbb{R}$ controls the gripper. A successful trajectory naturally supplies both labels: its recorded delta actions supervise f_{rel} , while its future end-effector states supervise f_{abs} . The geometric analysis and fusion operate on a_p ; although the absolute head also predicts gripper state during training, the deployed a_g always comes from the relative head.

B. Dual Action Manifolds

For an instruction l , let $\mathcal{M}_l \subset SE(3)$ denote the end-effector state manifold traced by successful demonstrations, and let TM_l denote its tangent bundle. Under nominal execution, the relative head predicts a local motion vector

$$\hat{\delta}a_{p,t} = f_{\text{rel}}(o_t, l, s_t)_p \in T_{p_t}\mathcal{M}_l, \quad (3)$$

where p_t is the current end-effector pose. This tangent representation is locally smooth and expressive, but it specifies only how to move from the current chart. Once a failure displaces the robot from the demonstrated support, the extrapolated vector field contains no explicit destination to which the robot should return.

The absolute head instead predicts a future coordinate

$$\hat{w}_{p,t} = f_{\text{abs}}(o_t, l)_p \in \mathcal{M}_l, \quad (4)$$

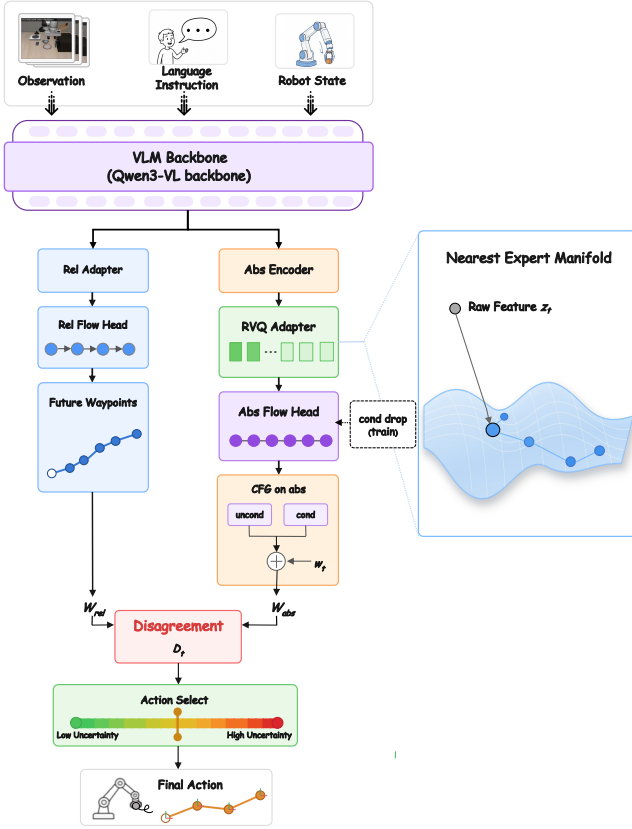


Fig. 2. DUET in four stages: a shared vision-language encoder, relative and absolute action heads, cross-space failure detection, and closed-loop recovery. The relative head provides state-aware local control; the explicitly state-free absolute head uses RVQ-anchored conditions to predict global waypoints. Their disagreement determines the transition between nominal execution and recovery.

which acts as a visual-language-conditioned global anchor. To compare the two representations, we introduce a state-dependent transport map

$$c_t = \Psi_{p_t}(\hat{w}_{p,t}) \in T_{p_t}SE(3), \quad (5)$$

which converts the absolute waypoint into a local correction at the live pose. Along an in-domain trajectory, c_t and $\delta a_{p,t}$ describe the same demonstrated motion in two coordinate systems and should therefore align. When a failure drives execution OOD, the relative head extrapolates a local field while the absolute head remains anchored to a demonstrated destination, exposing a cross-space inconsistency.

This geometry couples detection and recovery without asserting a formal OOD guarantee: the norm of the inconsistency provides a scalar failure signal, whereas the transported vector c_t already provides the corresponding recovery direction. DUET can therefore change both when and where the controller moves using the same pair of predictions.

C. Dual-Head Architecture

Figure 2 shows the shared VLM and two flow-matching action heads [16], [17]. The relative head attends to the full hidden sequence and explicitly receives s_t through a

numerical state encoder. It is therefore state-aware and serves as the primary controller during nominal execution. The absolute head receives no explicit current proprioception. When state tokens are appended to the causal VLM prompt, they are placed last and excluded from absolute-branch pooling, while the relative head can still attend to them. This separation prevents the absolute predictor from collapsing into an integration of the current state and preserves its role as a global reference.

The absolute branch pools its state-free hidden sequence H^0 and applies residual vector quantization (RVQ) [18], [19]:

$$Z_q = \text{RVQ}(\text{Pool}(H^0)). \quad (6)$$

RVQ maps continuous conditions to a discrete atlas learned from successful demonstrations, anchoring absolute predictions to recurrent visual-language modes rather than an unconstrained feature vector. Its residual norm is retained only as auxiliary telemetry and is not part of the primary failure score. The absolute flow head additionally uses classifier-free guidance (CFG) [20], enabling its conditional prediction to be softened toward the demonstrated waypoint marginal during recovery.

D. Disagreement-Preserving Learning

The two heads are supervised by the paired labels available in every successful trajectory. Both use flow-matching objectives, while RVQ contributes a commitment loss. The complete default objective is

$$\mathcal{L} = \lambda_{\text{rel}}\mathcal{L}_{\text{rel}} + \lambda_{\text{abs}}\mathcal{L}_{\text{abs}} + \lambda_{\text{vq}}\mathcal{L}_{\text{vq}}, \quad (7)$$

where the absolute labels are normalized future states and the relative labels are the recorded delta commands. We deliberately impose no cross-head consistency loss or distillation: forcing agreement everywhere would suppress the inconsistency needed for failure detection.

DUET instead specializes the branches through asymmetric visual augmentation. The absolute head and RVQ adapter train on both clean and perturbed observations, encouraging the global anchor to remain stable under degraded input. The relative head is optimized only on the clean subset $\mathcal{B}_{\text{clean}}$:

$$\mathcal{L}_{\text{rel}} = \frac{1}{|\mathcal{B}_{\text{clean}}|} \sum_{i \in \mathcal{B}_{\text{clean}}} \ell_{\text{FM}}(\hat{\Delta a}^{(i)}, \Delta a^{(i)}). \quad (8)$$

This gradient truncation preserves relative-head sensitivity while training the absolute branch as the robust reference, making cross-space disagreement more discriminative under visual OOD. For CFG training, the absolute head also replaces its quantized condition with learned null tokens on a subset of samples and learns both conditional and marginal waypoint flows.

E. Joint Failure Detection and Recovery

At deployment, the policy returns both unnormalized action chunks. Let the live pose be $p_t = (x_t, R_t)$ and the pose component of the current absolute waypoint be

$\hat{w}_{p,t} = (\hat{x}_t, \hat{R}_t)$. DUET implements the transport map in Eq. (5) as

$$c_t = \text{clip}\left(\left[\frac{(\hat{x}_t - x_t)/\sigma_p}{\text{Log}(\hat{R}_t R_t^{-1})/\sigma_R}\right], -1, 1\right), \quad (9)$$

where $\text{Log} : SO(3) \rightarrow \mathfrak{so}(3)$ gives the geodesic rotation correction and σ_p, σ_R convert pose error into action units. The cross-space disagreement is

$$D_t = \frac{\|c_t - \text{clip}(\hat{\delta}a_{p,t}, -1, 1)\|_2}{\sqrt{6}}. \quad (10)$$

This is the primary failure signal. We supplement it with a coarse state-distance term that measures normalized violation of the training-state bounds:

$$S_t = \text{mean}\left[\frac{\max(s_{\min} - p_t, 0)}{s_{\max} - s_{\min}} + \frac{\max(p_t - s_{\max}, 0)}{s_{\max} - s_{\min}}\right]. \quad (11)$$

The resulting failure score and recovery weight are

$$q_t = D_t + \lambda S_t, \quad (12)$$

$$\alpha_t = \sigma(k(q_t - \tau)). \quad (13)$$

DUET then interpolates between local execution and global recovery while retaining relative gripper control:

$$a_t = [a_{p,t}, a_{g,t}], \quad (14)$$

$$a_{p,t} = (1 - \alpha_t)\hat{\delta}a_{p,t} + \alpha_t c_t, \quad (15)$$

$$a_{g,t} = \hat{\delta}a_{g,t}, \quad (16)$$

where $[\hat{\delta}a_{p,t}, \hat{\delta}a_{g,t}]$ is the relative-head command. Thus relative control dominates along the demonstrated support, while a rising failure score shifts a_p toward the absolute recovery direction; control returns smoothly after the two representations realign. When requesting a new chunk, the previous fusion weight also softens the absolute-head CFG scale,

$$w_t = w_{\max} - (w_{\max} - w_{\min})\alpha_{t-1}. \quad (17)$$

The disagreement D_t therefore determines *when* recovery is needed, while the vector c_t determines *how* to recover.

IV. EXPERIMENTS

A. Comparison with Failure Detectors

Because DUET produces an explicit runtime failure score, we first compare its detection quality with representative detectors that require no failure examples for training. We implement logpZO, RND, and PCA-kmeans following the FAIL-Detect framework [4]; logpZO is the flow-based density score introduced by FAIL-Detect, RND adapts random network distillation [21] to observation–action novelty, and PCA-kmeans measures the distance from an observation embedding to clustered successful features. We additionally compare with STAC, which detects failures through temporal inconsistency between overlapping action predictions [9]. Table I reports discrimination, detection latency, and temporal localization under the same evaluation protocol.

DUET obtains the highest TPR and AUROC, indicating the strongest overall ranking of failed versus nominal execution in this comparison. Its lower TNR and longer detection delay reveal a complementary trade-off: DUET is more sensitive to failures, whereas RND rejects nominal alarms more effectively and PCA-kmeans triggers earlier. DUET also attains the second-highest Loc.@16 behind logpZO. We therefore treat detection quality as a separate empirical claim from the closed-loop retry results reported below.

B. Evaluation Questions and Protocol

Our evaluation follows the distinction made in the introduction between nominal execution, retryable deviations, and failures that invalidate the demonstrated behavior. We ask: **Q1: Nominal execution.** Does activating the dual-action controller retain standard task performance when no disturbance occurs? **Q2: Failure detection.** Does cross-action disagreement identify execution deviations with few nominal false alarms and low latency? **Q3: Trajectory re-entry.** When execution leaves the demonstrated path but that path remains reachable, does the detected disagreement improve task completion over relative control alone, and where does waypoint-based retry cease to apply? **Q4: Physical deployment.** Do the nominal-control and retry behaviors transfer to real manipulation across tasks with different contact structures?

We evaluate on the four LIBERO suites [22]—Spatial, Object, Goal, and LIBERO-10. Each suite contains 10 tasks, and each method is evaluated from 50 initial states per task, yielding 500 episodes per suite. We train one suite-specific dual-head checkpoint for 10k steps. *DUET* executes the adaptive controller in Eq. (16), whereas *DUET-Rel* fixes $\alpha_t = 0$ and executes only the relative head of the same checkpoint. This within-checkpoint ablation is our primary control: architecture, training data, optimization, task, initial state, and perturbation are held fixed, leaving the inference-time use of the absolute branch as the only difference. It isolates the controller, but not the training cost of adding the second head; the latter would require a separately trained, budget-matched single-head model.

For OOD evaluation, perturbations are scheduled at a fixed policy step with deterministic event seeds. DUET and DUET-Rel are therefore paired by identical (task, initial state) keys. We report closed-loop task success and paired McNemar p -values. We also report injection coverage whenever a perturbation has a semantic precondition. In particular, a grasp-drop event is counted as injected only if an object is actually held; episodes that do not satisfy this condition remain in the overall success rate and are not silently excluded.

The state-bound term S_t is nearly always zero in LIBERO. For the robot- and world-state experiments, we therefore use a residual-weighted disagreement score,

$$q_t = D_t r_t, \quad r_t = \text{clip}(\|c_{t,xyz}\|/0.05, 0, 1), \quad (18)$$

which suppresses a switch when the absolute prediction disagrees with the relative action but does not imply a

TABLE I

FAILURE-DETECTION COMPARISON. TPR, TNR, BALANCED ACCURACY (BAL.-ACC.), AUROC, AND LOC.@16 ARE HIGHER-IS-BETTER; DETECTION DELAY IS LOWER-IS-BETTER. BOLD VALUES INDICATE THE BEST RESULT IN EACH COLUMN.

Method	TPR $\uparrow$	TNR $\uparrow$	Bal.-Acc. $\uparrow$	AUROC $\uparrow$	Delay (steps) $\downarrow$	Delay (s) $\downarrow$	Loc.@16 $\uparrow$
DUET	0.900	0.265	0.582	0.845	38.3	1.91	0.523
logpZO [4]	0.889	0.243	0.566	0.817	22.7	1.14	0.569
RND [21], [4]	0.842	0.361	0.602	0.549	26.1	1.31	0.330
PCA-kmeans [4]	0.884	0.283	0.583	0.625	20.3	1.01	0.503
STAC [9]	0.839	0.311	0.575	0.494	25.8	1.29	0.260

TABLE II

NOMINAL TASK SUCCESS (%) ON LIBERO. EXTERNAL ROWS ARE REPORTED RESULTS AND ARE INCLUDED ONLY FOR BENCHMARK CONTEXT. STARVLA USES JOINT 30K TRAINING; DUET USES SUITE-SPECIFIC 10K CHECKPOINTS.

Method	Spatial	Object	Goal	LIBERO-10	Avg.
$\pi_{0.5}$ (reported)	98.8	98.2	98.0	92.4	96.9
StarVLA-GR00T (reported)	97.8	98.8	97.4	92.0	96.5
DUET	98.8	99.4	97.6	91.8	96.9
DUET-Rel	98.0	99.2	92.0	86.0	93.8

meaningful translational correction. Action-execution and contact experiments use the default score $q_t = D_t + \lambda S_t$. Camera blackout is handled separately because neither head can produce a meaningful prediction from an invalid image.

C. Nominal LIBERO Performance

Table II first tests whether failure-aware retry compromises ordinary execution. DUET attains 96.9% average success, matching the reported $\pi_{0.5}$ result and slightly exceeding the reported StarVLA-GR00T average. These external results provide benchmark context rather than a controlled comparison: StarVLA uses joint 30k training, whereas DUET uses suite-specific 10k checkpoints. The controlled comparison is DUET versus DUET-Rel. Activating adaptive fusion improves the average from 93.8% to 96.9%, with gains on all four suites. Thus, the retry mechanism does not trade away nominal task completion in this evaluation; a matched single-head retraining experiment is still needed to isolate any optimization effect of dual-head training itself.

D. OOD Detection and Failure-Aware Retry

Our OOD study compares only DUET with DUET-Rel. Reported clean scores from other methods are not valid OOD baselines because they do not share our initial states, injection times, severities, or success accounting. A fair comparison to another policy would require rerunning it under the identical paired protocol.

1) *Evaluation Scope and Perturbation Taxonomy*: We organize perturbations by where execution departs from successful demonstrations and, crucially, whether the demonstrated path remains a valid return target (Table III). Robot-state teleport and action-execution corruption displace the arm while preserving the task scene, so they directly test

DUET’s central claim: an absolute waypoint can guide the policy back to a reachable part of the learned behavior. A grasp drop instead changes the robot–object relation and can make the previous waypoint stale; it probes the boundary of waypoint-based retry rather than an unambiguous recovery case. External force is treated as an interaction stress test because it changes contact dynamics without necessarily creating a persistent pose error. Finally, camera blackout removes the input needed by both heads. It therefore requires an observation-validity guard and a cached waypoint, and should not be interpreted as evidence that disagreement alone detects arbitrary sensor failure.

Table IV summarizes the matched comparisons. We report each protocol over its own evaluation scope and avoid a grand average across heterogeneous disturbance types, severities, and horizons.

2) *Failure Detection Quality*: Closed-loop success alone cannot establish that disagreement is a useful failure detector: a controller could improve completion even if its score fires late or frequently activates during nominal execution. We therefore evaluate detection independently of recovery. Both heads are queried on trajectories executed by DUET-Rel, but their predictions do not affect the commanded action. This frozen-controller protocol evaluates every candidate signal on the same state sequence and prevents a successful correction from shortening its own positive example.

We construct event-level examples from the recoverable robot-state and action-execution shifts, for which the injection time is known and the demonstrated path remains valid. A positive example is a fixed window of H_{det} steps beginning at the actual injection time; its statistic is the maximum temporally smoothed detector value in that window. Negative examples are time-matched windows from clean rollouts with the same task and initial-state index. Initial states are partitioned into disjoint, task-stratified calibration and test sets. All smoothing parameters and thresholds are selected on the clean calibration split and frozen before evaluating perturbed episodes. Grasp drop and camera blackout are excluded: the former may invalidate the return target, while the latter uses an explicit observation-validity guard rather than disagreement.

We compare raw cross-action disagreement D_t with training-state bound violation S_t , RVQ residual, residual-weighted disagreement $D_t r_t$, and the default score $D_t + \lambda S_t$. Each signal and its calibrated threshold are applied

TABLE III

PERTURBATION TAXONOMY. THE FINAL COLUMN STATES THE ASSUMPTION UNDER WHICH AN ABSOLUTE WAYPOINT CAN SUPPORT TRAJECTORY RE-ENTRY.

Family	Shift	Protocol	What changes	Return-target status
Behavioral	Robot state	EEF teleport	Arm pose	Remains valid
	World state	Vertical grasp drop	Object/grasp relation	May become stale
	Action execution	Action perturbation	Command-to-motion map	Remains valid
Interaction	Contact dynamics	External force	Robot–world dynamics	Usually short-lived
Visual	Observation loss	Camera blackout	Current visual input	Cache last valid target

TABLE IV

CLOSED-LOOP TASK SUCCESS UNDER THE MATCHED OOD PROTOCOLS. EACH ROW AGGREGATES ONLY THE STATED SCOPE; HETEROGENEOUS SEVERITIES AND BLACKOUT HORIZONS ARE NOT POOLED INTO A SINGLE GRAND AVERAGE.

Family	Protocol	Scope	Pairs	DUET-Rel	DUET	Δ (pp)
Robot state	EEF teleport	4 suites, medium	2000	40.0	61.7	+21.7
World state	Vertical grasp drop	Object, medium	500	89.2	95.2	+6.0
Action execution	Action perturbation	4 suites, medium+severe	4000	27.9	64.6	+36.7
Interaction stress	External force	4 suites, medium+severe	4000	92.22	95.95	+3.73
Observation loss	Camera blackout	4 suites, step 15 medium	2000	83.4	92.2	+8.8
Observation loss	Camera blackout	4 suites, step 23 medium+severe	4000	73.5	90.8	+17.3

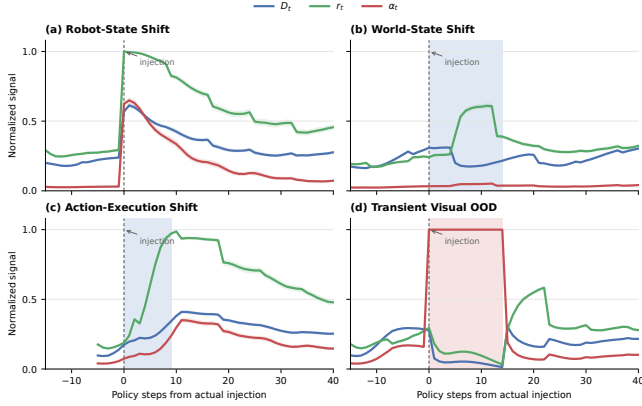


Fig. 3. Controller dynamics on the Object suite under (a) end-effector teleport, (b) vertical grasp drop, (c) action perturbation, and (d) camera blackout with a cached waypoint. Traces are aligned to the actual injection time and show means with pointwise bootstrap 95% confidence intervals over successfully injected episodes. Shading denotes the perturbation or forced-hold window.

unchanged to both perturbation families; selecting a score using the known shift label would leak protocol information into detection. Table V reports window-level AUROC and AUPRC, event recall at a threshold calibrated to 1% clean false-positive rate, median delay to the first threshold crossing among detected events, and false alarms per 1000 clean control steps. Confidence intervals are obtained by bootstrapping complete episodes rather than individual frames, preserving within-trajectory dependence.

3) Retry under Recoverable Behavioral Shifts:

a) *Robot-state shift.*: We first teleport the end effector away from its current pose while leaving the scene unchanged. This produces the failure mode targeted by DUET:

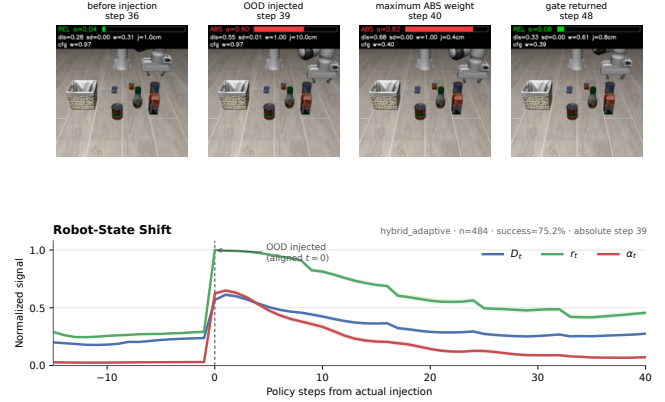


Fig. 4. End-effector teleport on the Object suite. Top: a successful retry, with relative control before the displacement, absolute waypoint correction immediately afterward, and a return to relative control after re-entry. Bottom: mean disagreement D_t , residual weight r_t , and fusion weight α_t over 484 successfully injected episodes, aligned to the injection time; bands are pointwise bootstrap 95% confidence intervals.

relative commands remain locally plausible around an off-trajectory state, whereas the absolute branch still identifies a reachable point on the demonstrated path. The residual factor r_t ensures that disagreement activates retry only when the predicted waypoint also requires a nontrivial pose correction.

The temporal response in Fig. 4 matches this interpretation. Immediately after the teleport, both D_t and r_t increase, driving α_t from near zero to above 0.6. As the arm approaches the predicted waypoint, the residual and gate decay, returning control to the relative branch. This switch-and-return behavior is also visible in Fig. 3(a); DUET does not remain in a permanent absolute-control mode.

TABLE V

FAILURE DETECTION ON FROZEN DUET-REL TRAJECTORIES. RECALL IS MEASURED AT A THRESHOLD CALIBRATED TO 1% CLEAN-WINDOW FPR; DELAY IS EVENT-LEVEL, AND FALSE ALARMS ARE MEASURED OVER FULL CLEAN ROLLOUTS. VALUES ARE PLACEHOLDERS UNTIL ANALYSIS IS COMPLETE.

Detection signal	End-effector teleport				Action perturbation				Clean
	AUROC	AUPRC	Recall	Delay	AUROC	AUPRC	Recall	Delay	FA/1k
State-bound violation S_t	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
RVQ residual	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
Cross-action disagreement D_t	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
Residual-weighted $D_t r_t$	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
Default score $D_t + \lambda S_t$	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO

TABLE VI

CLOSED-LOOP TASK SUCCESS (%) UNDER BEHAVIORAL PERTURBATIONS. DUET AND DUET-REL USE THE SAME CHECKPOINT. TELEPORT AND GRASP DROP USE RESIDUAL-WEIGHTED DISAGREEMENT; ACTION PERTURBATION USES THE DEFAULT SCORE.

Setting	Suite	DUET-Rel	DUET	Δ	p
<i>Robot-state shift</i> (end-effector teleport, medium, $N=500$)					
	Spatial	54.6	57.6	+3.0	0.21
	Object	36.2	76.0	+39.8	1×10^{-40}
	Goal	33.4	52.4	+19.0	2×10^{-12}
	LIBERO-10	35.8	60.8	+25.0	1×10^{-17}
	Overall	40.0	61.7	+21.7	—
<i>World-state shift</i> (vertical grasp drop, Object, $N=500$)					
	Object	89.2	95.2	+6.0	2.4×10^{-5}
<i>Action-execution shift</i> (action perturbation, step 5, $N=500$)					
medium	Spatial	54.0	74.0	+20.0	1×10^{-14}
medium	Object	48.8	84.4	+35.6	1×10^{-34}
medium	Goal	47.2	74.2	+27.0	5×10^{-22}
medium	LIBERO-10	36.6	78.6	+42.0	1×10^{-48}
severe	Spatial	12.8	43.2	+30.4	6×10^{-31}
severe	Object	7.8	60.6	+52.8	1×10^{-70}
severe	Goal	10.6	52.6	+42.0	3×10^{-49}
severe	LIBERO-10	5.4	49.0	+43.6	2×10^{-53}
	Overall	27.9	64.6	+36.7	—

Across 2000 paired episodes, DUET improves success from 40.0% to 61.7% (+21.7 percentage points; Table VI). The largest gain occurs on Object (+39.8 points), followed by LIBERO-10 (+25.0) and Goal (+19.0). Among Object episodes in which injection succeeds, DUET completes 72.8%, compared with 33.2% for DUET-Rel, directly measuring retry after the disturbance. The Spatial difference is smaller and not statistically significant (+3.0 points, $p=0.21$); roughly 12% of episodes terminate before the scheduled perturbation, and the remaining effect varies substantially by task. Failures also concentrate in contact-rich Goal tasks such as opening the top drawer and pushing a plate, for which a pose waypoint alone is an incomplete return target.

b) Action-execution shift. We next corrupt the executed 6D command over a short window. The task scene remains unchanged, but execution error accumulates over time rather than appearing as an instantaneous pose reset. Accordingly, disagreement and the fusion weight rise gradually in Fig. 3(c). This setting yields the largest improvement: over

eight suite-severity groups (4000 paired episodes), DUET reaches 64.6% success, compared with 27.9% for DUET-Rel (+36.7 points). Gains are consistent across all suites and both severities, and increase to +52.8 points on severe Object perturbations. Because the evaluator provides no perturbation flag to the policy, the switch is triggered by the two action predictions becoming inconsistent as the arm drifts from the learned trajectory.

4) Boundary Case: World-State Shift: A vertical grasp drop models a failed hold rather than a displacement of the arm. At policy step 67, if a free-joint object is in gripper contact, the evaluator opens the gripper, preserves the object’s XY position, lowers it by 5 cm, and keeps the gripper open for 15 steps. The event is marked `not_grasped` and the simulator is left unchanged when no object is held.

This intervention tests an important limit of DUET: the robot may remain near its demonstrated grasp pose even though the object has moved, so the previous absolute waypoint is not necessarily a valid target. On Object, DUET succeeds in 476/500 episodes (95.2%), compared with 446/500 (89.2%) for DUET-Rel, a +6.0-point difference (bootstrap 95% CI [3.4, 8.8]; McNemar $p=2.4 \times 10^{-5}$). Injection succeeds in 498/500 DUET episodes and 494/500 DUET-Rel episodes. However, Fig. 3(b) shows that the mean fusion weight remains low during the forced-open interval. The aggregate evidence therefore does not support interpreting this gain as broad absolute-waypoint recovery; instead, it shows that residual weighting avoids aggressively returning to a potentially stale grasp while retaining a modest completion benefit.

Coverage is lower on Goal because many tasks do not involve a grasp at the fixed injection time. Both methods receive the drop in only 282/500 paired episodes. Overall success is 53.6% for DUET and 50.4% for DUET-Rel; among the both-injected subset it is 23.4% and 17.7%, respectively. Because this difference is dominated by one bowl-to-stove task, we report it as a coverage diagnostic rather than a suite-level world-state result.

5) Interaction Stress Test: We additionally apply an external force during contact, perturbing robot-world dynamics without directly resetting the end-effector pose. Across eight suite-severity groups, DUET reaches 95.95% success versus 92.22% for DUET-Rel (+3.73 points; Table VII).

TABLE VII

EXTERNAL-FORCE INTERACTION STRESS TEST (FIXED-STEP, 100% INJECTION, $N=500$ PER CELL).

Suite	Medium			Severe		
	DUET-Rel	DUET	Δ	DUET-Rel	DUET	Δ
Spatial	96.6	98.4	+1.8	96.8	96.8	0.0
Object	97.6	99.0	+1.4	96.2	96.8	+0.6
Goal	89.2	96.8	+7.6	87.2	96.8	+9.6
LIBERO-10	87.8	92.6	+4.8	86.4	90.4	+4.0
Overall	DUET-Rel 92.22	DUET 95.95	$\Delta = +3.73$			

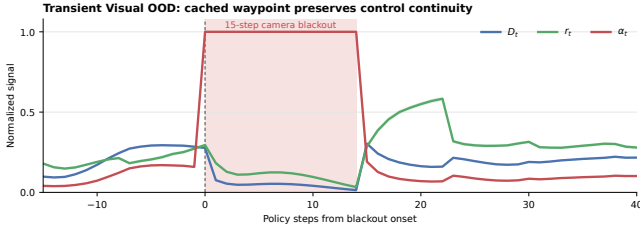
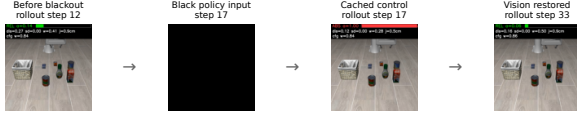


Fig. 5. Camera-blackout handling. An observation-validity guard skips inference on black frames; DUET follows the last reliable waypoint with $\alpha_t = 1$ and resumes the adaptive gate when vision returns. The lower panel aggregates 500 Object-suite episodes aligned to blackout onset.

The difference is non-negative in every cell and largest on Goal (+7.6 and +9.6 points), where drawer and plate interactions amplify short contact disturbances. The smaller margin relative to teleport and action corruption is consistent with a force impulse producing a weaker or shorter-lived waypoint residual. We therefore treat this experiment as evidence of interaction robustness, not as a separate claim of task-level recovery.

6) *Transient Observation Loss*: A complete camera blackout differs from the behavioral shifts above: the current observation is invalid, so disagreement between predictions on that image is not a meaningful failure signal. Indeed, querying both heads on black frames reduces success to 37.7%, below the 73.5% of DUET-Rel on the eight-group step-23 protocol. We therefore use a simple observation-validity guard. During blackout, DUET skips policy inference, corrects toward the last reliable absolute waypoint with $\alpha_t = 1$, and resumes adaptive control as soon as vision returns. This experiment evaluates whether the global waypoint provides a useful short-term execution memory; it does not claim endogenous detection of arbitrary visual corruption.

Figures 3(d) and 5 confirm that the cached waypoint is active only during the blackout. On the matched step-15 medium protocol, it improves success from 83.4% to 92.2% (+8.8 points; Table VIII). The largest gain is on LIBERO-10 (+22.4 points), while Object remains at ceiling. Under the longer step-23 protocol, the gap widens to +17.3 points

TABLE VIII

TASK SUCCESS (%) UNDER CAMERA BLACKOUT. DUET HOLDS THE LAST RELIABLE ABSOLUTE WAYPOINT. STEP-15 MEDIUM USES 2000 PAIRED EPISODES; STEP-23 AGGREGATES EIGHT SUITE-SEVERITY GROUPS.

Protocol	Suite	DUET-Rel	DUET	Δ	p
<i>Step 15, medium ($N=500$)</i>					
	Spatial	91.2	96.8	+5.6	3×10^{-5}
	Object	97.4	97.0	-0.4	0.84
	Goal	80.8	88.2	+7.4	5×10^{-4}
	LIBERO-10	64.2	86.6	+22.4	1×10^{-17}
	Overall	83.4	92.2	+8.8	—
<i>Step 23, medium+severe ($N=4000$)</i>					
	Overall	73.5	90.8	+17.3	—

(90.8% versus 73.5%). The main counterexample is the Goal task “open the top drawer and put the bowl inside,” where DUET is 32 points worse at step 15: a cached pre-blackout pose cannot complete a contact sequence that still requires updated visual feedback.

E. Real-Robot Evaluation

a) *Platform and protocol*.: We deploy DUET on a Franka manipulator and evaluate three tasks with distinct control requirements (Fig. 6). *Pick-and-place* requires free-space reaching, grasping, and object transport; *drawer opening* requires precise handle acquisition followed by constrained contact motion; and *whiteboard wiping* requires sustained surface contact while tracking an extended path. The final manuscript will additionally report the Franka model, end-effector, camera placement, control frequency, number of demonstrations per task, training budget, and end-to-end inference latency. As in simulation, training uses only successful demonstrations, without failure labels or human takeover trajectories.

For each task, we evaluate nominal execution and a controlled end-effector displacement that leaves the scene and task objective unchanged. The displacement is injected before grasp acquisition for pick-and-place, during the final handle approach for drawer opening, and during the contact stroke for whiteboard wiping. This common intervention isolates trajectory re-entry while exposing the controller to free-space, constrained, and sustained-contact regimes. The task-specific displacement direction and magnitude are fixed before evaluation and measured from Franka forward kinematics rather than estimated from video.

Injection occurs only after a predefined visual or contact precondition and must remain within prespecified workspace and force limits. A trial is marked as injected when these criteria are satisfied at intervention time; no trial is excluded based on its subsequent outcome. Trials that never reach the precondition are reported separately as uninjected rather than counted as retry failures. Scene resets use physical markers or fixtures, and DUET and DUET-Rel trials are interleaved in randomized blocks to reduce confounding from hardware drift. Outcomes are scored using predefined task criteria:

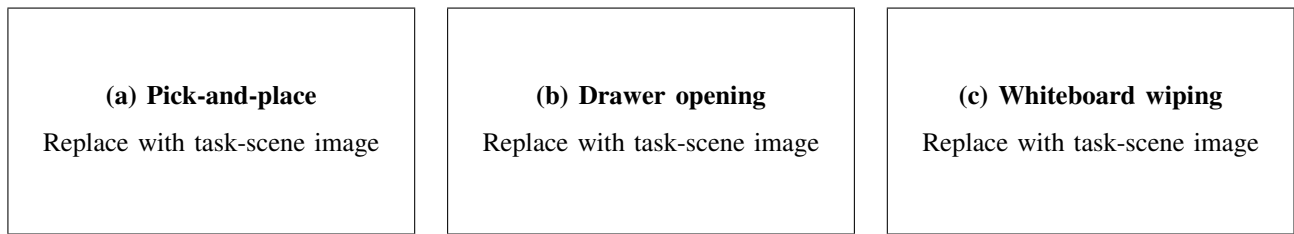


Fig. 6. Reserved layout for the Franka real-robot setup and three evaluation scenes: (a) pick-and-place, (b) drawer opening, and (c) whiteboard wiping. The final panels should mark the disturbance direction and injection phase so that the recoverability assumption is visually explicit.

- **Pick-and-place:** the commanded object is lifted and deposited fully inside the target receptacle.
- **Drawer opening:** the robot acquires the correct handle and moves the drawer beyond a prespecified displacement threshold.
- **Whiteboard wiping:** the robot maintains safe contact and removes at least a prespecified fraction of a standardized marked region.

In addition to overall task success, we report injection coverage and *retry success*: completion among trials satisfying the prespecified injection criteria. We report successes/trials with Wilson 95% confidence intervals rather than hiding the small physical-trial denominator behind percentages. A hardware safety stop is always active but is not used as DUET’s failure signal; safety-stop events are reported as failures.

F. Discussion

The evaluation separates four claims that are otherwise easy to conflate. Nominal runs test whether the additional control mode is safe to invoke; frozen-controller detection evaluates whether disagreement is a reliable alarm; matched OOD rollouts test whether the resulting correction improves task outcomes; and the Franka experiments test whether these behaviors survive physical contact and deployment noise. This separation prevents a higher success rate from being treated, by itself, as evidence of accurate failure detection.

The completed simulation results support the specific retry claim made in the introduction. DUET retains nominal LIBERO performance (96.9% average) rather than sacrificing local control for global correction. When the demonstrated path remains a valid target, the same-checkpoint comparison shows substantial gains under end-effector teleport (+21.7 points) and action corruption (+36.7 points). The controller traces provide complementary mechanism evidence: disagreement and the waypoint residual increase after deviation, absolute correction becomes dominant, and relative control resumes after re-entry. The detection metrics in Table V and physical results in Table IX must be completed before making quantitative claims about alarm quality or real-world transfer.

The remaining experiments delimit rather than broaden this claim. A grasp drop can invalidate the old world reference, and DUET correspondingly keeps the average gate low; external force often induces only a short-lived correction; and complete observation loss requires an explicit validity guard plus a cached waypoint. DUET should therefore be

interpreted as a failure-aware retry controller for reachable deviations from learned behavior, not as a general planner for irreversible failures or tasks that require a new strategy.

V. LIMITATIONS

DUET assumes access to absolute state labels aligned with the relative action chunks. This is natural in LIBERO and many robot logs, but requires accurate calibration and state estimation on the Franka platform. The absolute waypoint is an RVQ demonstration anchor, not an object tracker: a world-state shift can invalidate the grasp reference even when the robot remains nearby. The bound-violation term S_t is uninformative on LIBERO, so robot-state gating relies on the residual to the predicted waypoint; this still cannot distinguish “the arm left a valid target” from “the target itself moved.” Complete visual dropout requires an explicit validity guard and a cached waypoint rather than disagreement-based detection. Matched component ablations, detection analysis, and Franka results remain incomplete in this draft.

VI. CONCLUSION

We presented DUET, a dual-branch VLA policy for endogenous failure detection and trajectory re-entry. By aligning local relative actions with stateless, RVQ-anchored absolute waypoints, DUET uses cross-space disagreement to determine when execution has left learned behavior and uses the waypoint correction to retry it. The resulting controller preserves nominal performance and substantially improves completion under recoverable pose and action-execution deviations without failure labels or takeover demonstrations. Its scope is deliberately narrower than general recovery planning: stale world references, irreversible failures, and fully invalid observations require additional mechanisms or a new task strategy.

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TABLE IX

REAL-ROBOT EVALUATION. SUCCESS ENTRIES ARE REPORTED AS SUCCESSES/TRIALS (%); RETRY SUCCESS CONDITIONS ONLY ON TRIALS MEETING THE PRESPECIFIED INJECTION CRITERIA. VALUES AND DISTURBANCE DEFINITIONS ARE PLACEHOLDERS PENDING PHYSICAL EXPERIMENTS.

Task	Recoverable disturbance	Coverage	Nominal success		Disturbed success		Retry success		
			DUET-Rel	DUET	DUET-Rel	DUET	DUET-Rel	DUET	Δ
Pick-and-place	EEF shift before grasp (TODO cm)	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
Drawer opening	EEF shift before contact (TODO cm)	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
Whiteboard wiping	EEF shift during stroke (TODO cm)	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO
Overall	—	TODO	TODO	TODO	TODO	TODO	TODO	TODO	TODO

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